

# SAT Hard Question Set 7

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June 18, 2026

# Novel Prep



# Overview



# Question 1

A rectangular region of land is divided into 54 square lots of equal area  $S$  in square units. The length of the region is 1.5 times the width of the region. If the width of the region is  $a\sqrt{S}$  units, what is the value of  $a$ ?

- A. 5
- B. 6
- C. 7
- D. 8

# Answer 1

**Correct Answer: B**

## Question 2

In the given pair of equations,  $a$  and  $b$  are constants. The graph of this pair of equations in the  $xy$ -plane is a pair of perpendicular lines.

Which of the following pairs of equations also represents a pair of perpendicular lines?

$$5x + 7y = 1$$

$$ax + by = 1$$

A.  $10x + 7y = 1$ ,  $ax - 2by = 1$

B.  $10x + 7y = 1$ ,  $ax + 2by = 1$

C.  $10x + 7y = 1$ ,  $2ax + by = 1$

D.  $5x - 7y = 1$ ,  $ax + by = 1$

# Answer 2

**Correct Answer: B**



## Question 3

A real estate company offers a series of three webinars. 2,000 people attended the first webinar. 65% of the people who attended the first webinar attended the second webinar, and 44% of the people who attended both the first and second webinars attended the third webinar.

If those who attended the first but did not attend the second webinar, 31% attended the third webinar.

How many people attended both the first and third webinars but did not attend the second webinar?

- A. 215
- B. 217
- C. 219
- D. 220

# Answer 3

**Correct Answer: B**

## Question 4

The functions  $f$  and  $g$  are defined by the given equations, where  $x \geq 0$ :

I  $f(x) = 19(1.27)^x + 43$

II  $g(x) = 8(0.77)^x$

Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where  $x \geq 0$ ?

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

# Answer 4

**Correct Answer: B**

# Question 5

Circle  $F$  in the  $xy$ -plane is represented by the equation:

$$(x - 13)^2 + (y - 9)^2 = 196$$

Circle  $G$  is obtained by shifting circle  $F$  5 units to the left and 11 units up. An equation representing circle  $G$  is:

$$(x + h)^2 + (y + k)^2 = 196$$

where  $h$  and  $k$  are constants.

What is the value of  $h + k$ ?

# Answer 5

**Final Answer:**  $-28$

# Question 6

The function  $q$  is defined by:

$$q(x) = a|x - 9|^2 - 87|x - 9| + b$$

where  $a$  and  $b$  are constants, and  $a > b > 87$ .

If  $q(639) = h$  and  $q(-621) = k$ , where  $h$  and  $k$  are constants, what is the value of:

$$9(-639)^{h-k} + 621(9)^{k-h}?$$

# Answer 6

**Final Answer:** 630



# Question 7

In the given equation,  $a$  and  $b$  are positive constants. A solution to the equation is  $x = 42$ .

$$x(9x - 2a) + b(9x - 2a) - (x + b) = 0$$

What is the value of  $a$ ?

# Answer 7

**Final Answer:** 188.5

## Question 8

A right rectangular prism has a base area of  $28p$  square centimeters ( $\text{cm}^2$ ). The length of the base of the rectangular prism is 12 cm, and the height of the rectangular prism is 4 cm.

Which expression represents the surface area, in  $\text{cm}^2$ , of the right rectangular prism?

- A.  $\frac{512p}{3}$
- B.  $\frac{224p}{3} + 96$
- C.  $56p + 192$
- D.  $560p + 96$

# Answer 8

**Correct Answer:** B

# Question 9

The expression

$$\frac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$$

where  $x > 1$  and  $y > 1$ , is equivalent to which of the following?

- A.  $\frac{\sqrt{y}}{\sqrt[3]{x}}$
- B.  $\frac{y\sqrt{y}}{\sqrt[3]{x}}$
- C.  $\frac{y\sqrt{y}}{x\sqrt[3]{x}}$
- D.  $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

# Answer 9

**Correct Answer: D**

# Question 10

The function  $z$  is defined by the equation:

$$z(w) = (0.829)^{2w}$$

The value of  $z(w)$  decreases by  $p\%$  for each increase by 1 in the value of  $w$ . Which of the following is closest to the value of  $p$ ?

- A. 0.171
- B. 0.829
- C. 17.1
- D. 31.3

# Answer 10

**Correct Answer: D**



# Question 11

$$-16(5x - 3)^2 + 4(5x - 2)^2$$

The given expression can be rewritten as:

$$\frac{a}{4}x^2 + \frac{b}{4}x + \frac{c}{4}$$

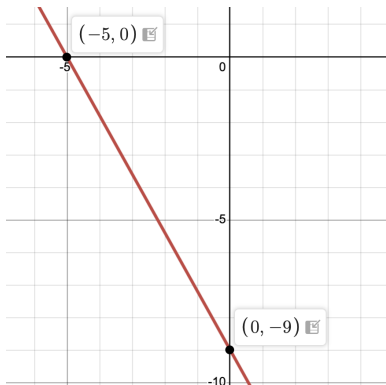
where  $a$ ,  $b$ , and  $c$  are constants. What is the value of  $a + b + c$ ?

- A. -112
- B. -56
- C. -28
- D. -7

# Answer 11

**Correct Answer: A**

# Question 12



The graph of line  $h$  is shown in the  $xy$ -plane. Line  $k$  (not shown) is defined by the equation:

$$sx + 40y = t$$

where  $s$  and  $t$  are constants. If line  $k$  is graphed in this  $xy$ -plane, the result is a graph of two linear equations. This system of two linear equations has no solution. Which of the following is **NOT** a possible value of  $t$ ?

# Answer 12

**Correct Answer: A**

# Thank You!



We appreciate your time and focus.

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